

Smart Complaint Management System

Software Requirements Specification — Lab 1

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1. User Stories

ID	Role	User Story	Prior ity	Acceptance Criteria
US-01	Citizen	As a citizen, I want to submit a complaint with a photo and GPS location so that the concerned department knows exactly what and where the issue is.	Must	Complaint saved with image and map pin; confirmation ID shown.
US-02	Citizen	As a citizen, I want to track the real-time status of my complaint so that I know what stage it is in without calling a helpline.	Must	Timeline view shows status transitions with timestamps; push notification on each change.
US-03	Citizen	As a citizen, I want to rate the resolution and leave feedback so that service quality can be measured.	Must	1-5 star rating and comment saved; CSAT score computed.
US-04	Citizen	As a citizen, I want to receive SMS and email updates so that I stay informed even without the app.	Should	SMS sent on status changes for opted-in users; email with complaint summary.
US-05	Agent	As an agent, I want to see complaints routed to my department sorted by SLA urgency so that I handle the most critical ones first.	Must	Dashboard shows SLA countdown; overdue items highlighted at the top.
US-06	Agent	As an agent, I want to add internal notes and attach evidence to a complaint so that the resolution trail is documented.	Must	Notes saved with agent name and timestamp; attachments support PDF and images up to 10 MB.
US-07	Agent	As an agent, I want the system to suggest a category automatically so that I spend less time on manual triage.	Should	NLP model suggests top-3 categories with confidence scores above 70%.
US-08	Supervisor	As a supervisor, I want to escalate a complaint to another team so that stalled tickets get expert attention.	Must	Escalation transfers ticket to priority queue with a new SLA; all parties notified.
US-09	Supervisor	As a supervisor, I want a dashboard of KPIs (open, resolved, overdue, avg time) so that I can monitor team performance.	Must	Dashboard refreshes every 60 s; export to CSV available.
US-10	Admin	As an administrator, I want to configure SLA rules per category and severity so that response expectations match organisational policy.	Must	SLA rules editable via form; changes effective on new complaints immediately.

2. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow citizens to submit complaints via web or mobile, including title (max 120 chars), description (max 2000 chars), category, severity, up to 5 images (max 5 MB each), and optional GPS coordinates.	High	Pass: Complaint saved with all fields. Fail: Mandatory field missing or oversized attachment accepted.	Core submission workflow
FR-002	Functional	The system shall use an NLP model to suggest top-3 complaint categories with confidence scores; citizen may accept or override.	High	Pass: Suggestions shown with $\geq 70\%$ confidence. Fail: No suggestion or wrong category ranked first.	Reduces manual triage effort
FR-003	Functional	The system shall generate a unique ticket ID (SCMS-YYYYMMDD-NNNNN) on successful submission.	High	Pass: Unique ID shown on confirmation. Fail: Duplicate ID or missing.	Tracking reference for all parties
FR-004	Functional	The system shall route new complaints to the appropriate department and agent based on category, location, and agent workload.	High	Pass: Complaint lands in correct dept queue. Fail: Routed to wrong department.	Eliminates manual assignment
FR-005	Functional	The system shall enforce a complaint lifecycle: New -> Assigned -> In Progress -> Pending Info -> Resolved -> Closed, logging every transition with timestamps.	High	Pass: Status history shows all transitions. Fail: Transition skipped or unlogged.	Auditability and transparency
FR-006	Functional	The system shall compute SLA deadlines per complaint based on category and severity rules, and trigger alerts when approaching (< 2 h) or breached.	High	Pass: Alerts sent at warning and breach thresholds. Fail: No alert on SLA breach.	Accountability for resolution times
FR-007	Functional	The system shall send notifications via in-app, email, and SMS (opt-in) on status changes, SLA warnings, and resolution confirmation.	Medium	Pass: Notification received within 60 s. Fail: Notification missing or delayed > 5 min.	Keeps citizens informed
FR-008	Functional	Upon resolution, the system shall prompt the citizen for a 1-5 star rating and optional comment within 7 days; unrated complaints auto-close.	Medium	Pass: Rating saved and CSAT updated. Fail: Rating not recorded.	Measures service quality
FR-009	Functional	The system shall allow supervisors to escalate complaints, resetting SLA timers and moving tickets to a priority queue.	High	Pass: Ticket moves to priority queue with new SLA. Fail: SLA not reset.	Handles stalled complaints
FR-010	Functional	The system shall support CRUD on user accounts with RBAC for four roles: Citizen, Agent, Supervisor, Admin.	High	Pass: Permissions enforced per role on every API call. Fail: Unauthorised access succeeds.	Security and access control

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-011	Functional	The system shall provide real-time dashboards showing open/closed counts, avg resolution time, SLA compliance rate, and CSAT trends.	Medium	Pass: Dashboard refreshes every 60 s with accurate data. Fail: Stale metrics.	Performance monitoring
FR-012	Functional	The system shall maintain an immutable audit log of every complaint action with actor ID and timestamp.	High	Pass: Full action history retrievable. Fail: Any action unlogged.	Regulatory compliance
NFR-001	Nonfunctional	The system shall handle 500 concurrent users with API response times under 500 ms (p99) and page loads under 2 seconds (p95).	High	Pass: Load test confirms metrics. Fail: Latency exceeds thresholds.	Responsive user experience
NFR-002	Nonfunctional	All data in transit shall use TLS 1.3; data at rest encrypted with AES-256. OWASP Top 10 mitigated.	High	Pass: Security audit confirms compliance. Fail: Any PII found unencrypted.	Data protection and privacy
NFR-003	Nonfunctional	The system shall maintain 99.9% uptime (< 8.76 h downtime/year). RPO: 1 hour, RTO: 30 minutes.	High	Pass: Uptime report confirms target. Fail: Downtime exceeds threshold.	Service reliability

3. Test Cases

Test cases follow the template: Test Case ID, Name of Module, Test case description, Pre-conditions, Test Steps, Test data, Expected Results, Actual Result, Test Result.

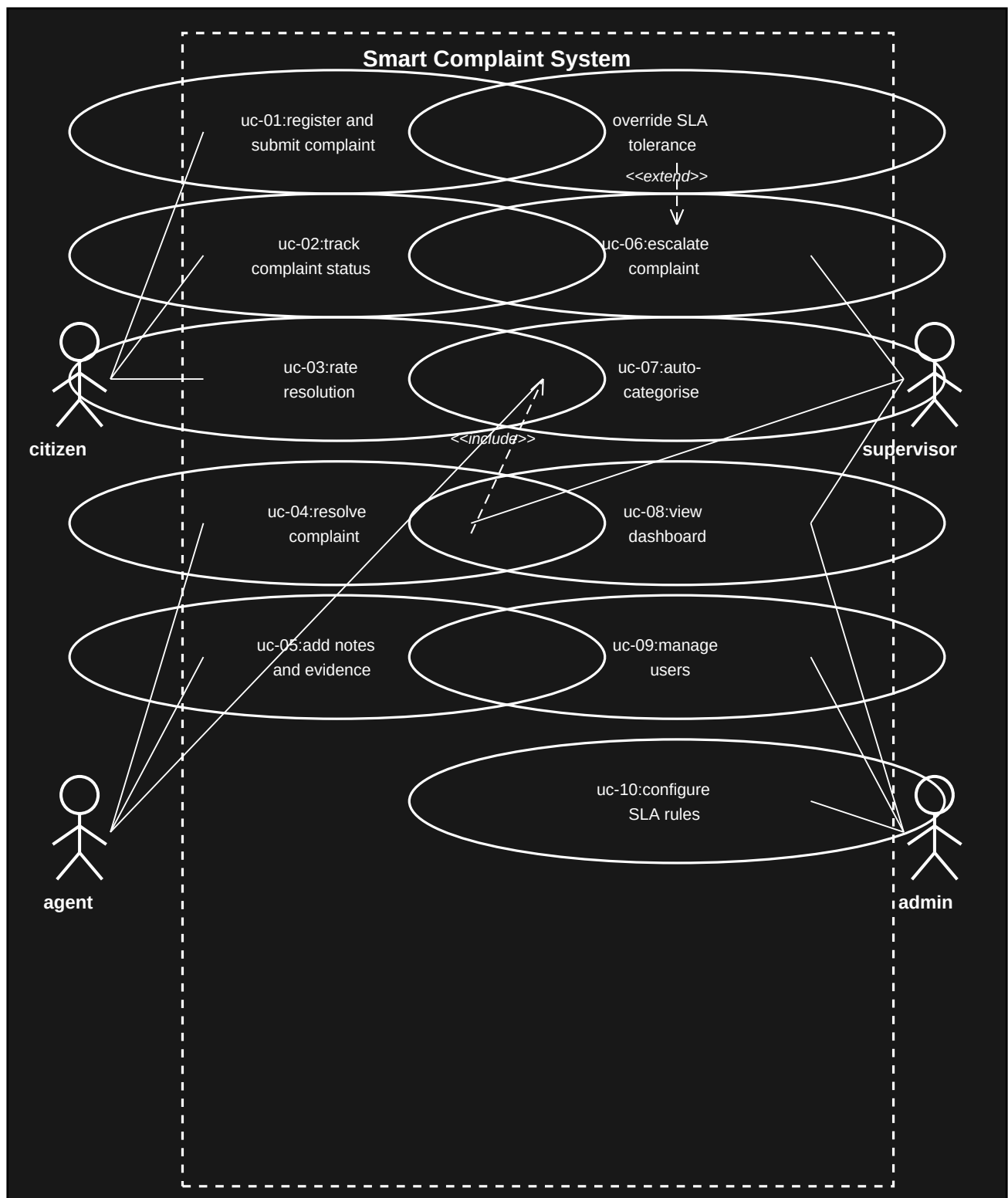
Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-01	Complaint Submission	To test the complaint form submission with valid inputs	Citizen is logged in; on home screen	1: Click 'New Complaint' 2: Enter title 'Pothole on MG Road' 3: Enter description 4: Select category 'Roads' 5: Set severity 'High' 6: Attach image (2 MB) 7: Enable GPS 8: Click Submit	Title: Pothole on MG Road Category: Roads Severity: High Image: pothole.jpg (2 MB)	Complaint created with ticket ID SCMS-YYYYMMDD-NNNNN; confirmation screen with timeline link	Complaint created successfully with ID SCMS-20260902-00001; confirmation displayed	Pass
UT-02	Complaint Submission	To test form validation when mandatory fields are empty	Citizen is logged in	1: Click 'New Complaint' 2: Leave title empty 3: Leave category unselected 4: Click Submit	Title: (empty) Category: (none) Severity: (none)	Validation error: 'Title is required', 'Category is required' shown; form not submitted	Validation errors displayed correctly; form not submitted	Pass
UT-03	Complaint Submission	To test rejection of oversized attachment	Citizen is logged in; filling complaint form	1: Click 'Attach Image' 2: Select a 12 MB file 3: Observe response	Image: largefile.jpg (12 MB)	Error message: 'File exceeds 5 MB limit'; file not attached	Error message displayed; file rejected	Pass
IT-01	Auto-Categorisation	To test NLP model suggests correct category for complaint text	NLP model deployed and active	1: Citizen types 'water leakage from main pipe on 5th cross' 2: System processes text 3: Observe suggested categories	Description: 'water leakage from main pipe on 5th cross'	Top suggestion: 'Water Supply' with confidence $\geq 70\%$; top-3 list shown	Top suggestion: Water Supply (87%); list shown correctly	Pass
IT-02	Smart Routing	To test complaint routes to correct department and agent	Complaint submitted under category 'Water Supply' in Zone 3; agents active	1: New complaint created 2: System evaluates routing rules 3: Observe assignment	Category: Water Supply Location: Zone 3	Complaint assigned to Water Dept - Zone 3 agent with lowest current workload	Assigned to Agent Ravi (Water Dept - Zone 3, 4 active tickets)	Pass

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT-03	SLA Engine	To test SLA breach triggers alert notifications	Complaint with 24 h SLA created; 22 hours elapsed	1: Wait for SLA warning threshold (2 h remaining) 2: Check agent notifications 3: Wait for SLA breach (0 h) 4: Check supervisor notifications	SLA: 24 h Elapsed: 22 h	Amber alert to agent at 2 h remaining; red alert to supervisor at breach	Amber alert sent to agent; red alert sent to supervisor at breach	Pass
ST-01	Complaint Lifecycle	To test full complaint lifecycle from submission to closure	Citizen and Agent are logged in; system active	1: Citizen submits complaint 2: System assigns to agent 3: Agent changes status to 'In Progress' 4: Agent adds resolution note 5: Agent marks 'Resolved' 6: Citizen rates 4 stars 7: Complaint auto-closes	Valid complaint data; rating: 4 stars; comment: 'Fixed well'	Status transitions: New -> Assigned -> In Progress -> Resolved -> Closed; all logged with timestamps	All status transitions logged correctly; CSAT updated	Pass
ST-02	Escalation	To test supervisor escalation of stalled complaint	Complaint in 'In Progress' for > 48 h; supervisor logged in	1: Supervisor opens overdue tab 2: Selects stalled complaint 3: Clicks 'Escalate' 4: Enters reason 5: Selects senior team 6: Confirms	Complaint ID: SCMS-20260830-00042 Reason: 'No response from field team'	Ticket moved to priority queue; new SLA timer starts; original agent and citizen notified	Ticket escalated; new SLA set; notifications sent to all parties	Pass
ST-03	User Management (RBAC)	To test admin creates agent account with correct role permissions	Admin is logged in	1: Navigate to User Management 2: Click 'Add User' 3: Enter name, email, department 4: Select role 'Agent' 5: Click Save	Name: Priya S Email: priya@scms.in Dept: Roads Role: Agent	Account created; welcome email sent; agent can log in and see only assigned complaints	Account created; welcome email sent; agent logged in with correct permissions	Pass
ST-04	Security	To test SQL injection prevention on complaint title field	System running; citizen logged in	1: In complaint title, enter: ' OR 1=1 -- 2: Fill remaining fields normally 3: Click Submit	Title: ' OR 1=1 -- Other fields: valid data	Input sanitised; no SQL execution; complaint saved with escaped characters	Input sanitised; complaint saved normally with escaped text	Pass

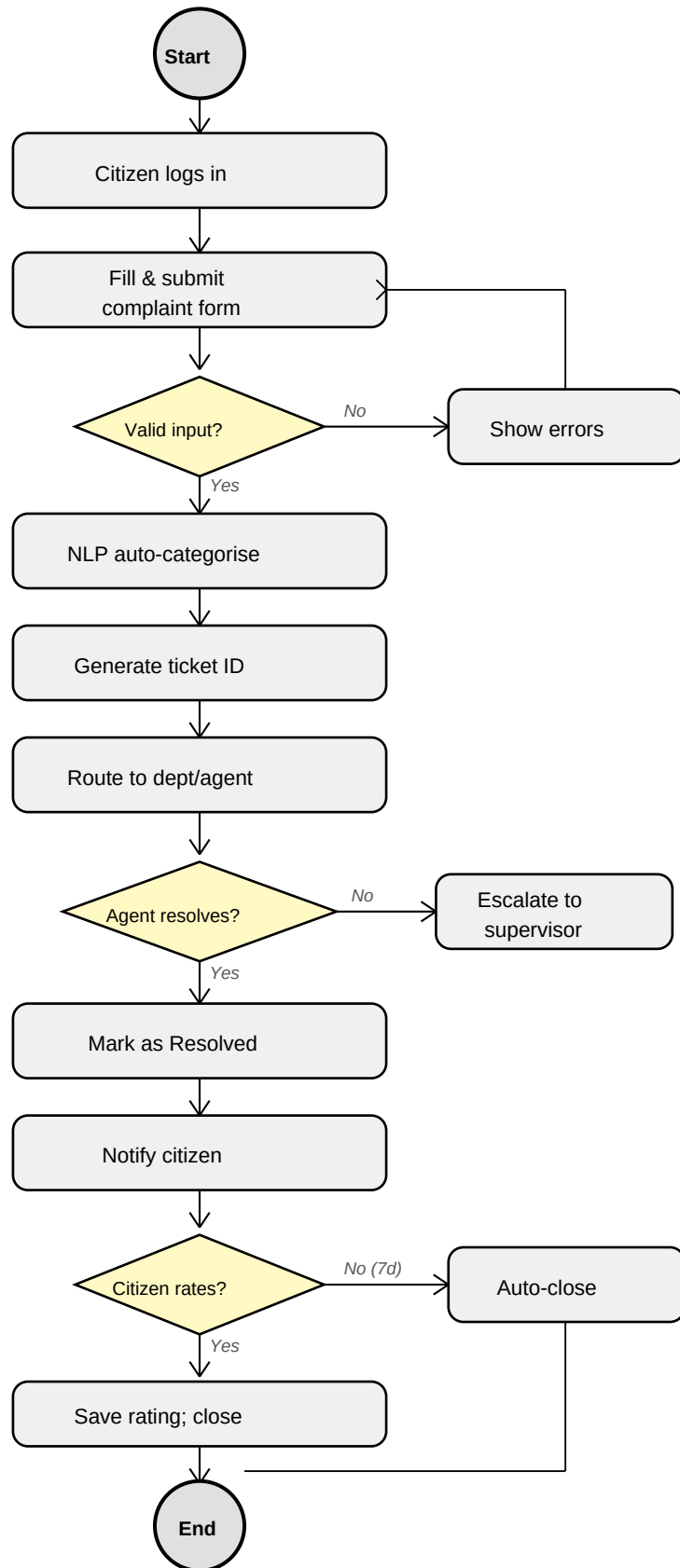
4. UML Use-Case Diagram

Actors: Citizen, Agent, Supervisor, Admin

Relationships: UC-04 (Resolve Complaint) <> UC-07 (Auto-Categorise); Escalate Complaint <> UC-04 (Resolve Complaint)



4b. Workflow Diagram: Submit & Resolve Complaint



5. Use-Case Flow: UC-01 Submit Complaint

Primary Actor: Citizen

Secondary Actors: Agent, NLP Service

Description: Citizen files a new complaint; system validates, categorises, routes it to the correct department, and confirms.

Preconditions:

- Citizen is logged in and on the home screen.
- At least one agent is active in the target department.

Postconditions:

- A new complaint ticket is created, stored, and routed to the appropriate department.
- Citizen receives a confirmation notification with the ticket ID.

Main Success Scenario

1. Citizen taps "New Complaint". System shows the submission form.
2. Citizen fills in the title, description, and selects severity.
3. System runs NLP auto-categorisation and suggests top-3 categories with confidence scores.
4. Citizen accepts or overrides the suggested category.
5. Citizen optionally attaches up to 5 images (max 5 MB each) and enables GPS location.
6. Citizen taps "Submit".
7. System validates all mandatory fields (title, description, category, severity).
8. System generates a unique ticket ID (SCMS-YYYYMMDD-NNNNN).
9. System routes the complaint to the appropriate department/agent using smart routing rules (category + location + agent workload).
10. System sends confirmation notification (in-app + email) with ticket ID and expected SLA.
11. System displays confirmation screen with ticket details and a timeline link.

Alternate Flow

3a. NLP Confidence Too Low

- 3a.1. All suggestions have confidence < 70%.
- 3a.2. System asks citizen to manually select a category from the full list.
- 3a.3. Citizen selects a category; flow resumes at step 4.

7a. Validation Fails

- 7a.1. System highlights missing mandatory fields with error messages.
- 7a.2. Citizen corrects the fields and resubmits.
- 7a.3. Flow resumes at step 7.

5a. Attachment Exceeds Limit

- 5a.1. System rejects the file and displays 'File exceeds 5 MB limit'.
- 5a.2. Citizen removes or replaces the attachment.
- 5a.3. Flow resumes at step 5.

9a. No Agent Available

9a.1. No agent is currently active in the target department.

9a.2. System assigns complaint to the supervisor queue and sends an escalation alert.

9a.3. Flow resumes at step 10.